3 Sets

Introduction to Sets

Definition: A set is an unordered collection of "objects". The *objects* in a set are called the elements or members of the set. A set contains its elements. A set is usually denoted by a capital letter, say A, and an element is usually denoted by a lowercase letter, say a or b. We write $a \in A$ to mean "a is an element of A". $a \notin A$ means the statement $\sim(a \in A)$, or "a is not an element of A".

For example

- consider the objects 0, 2, -3, 5. The collection of objects 0, 2, -3, 5 is a set and we *denote this set* by, {0, 2, -3, 5}. 0, 2, -3, 5 are *the members* of this set, i.e. they are the only members of this set.
- The fact that -3 is a member of the set {0, 2, -3, 5 } .
- 4 is not a member of this set.

$A = \{0, 2, -3, 5\}$

$-3 \in A$

$-4 \notin A$

Note:

When we consider the collection of all integers we get a set. Any integer is a member of this set. Any object which is not an integer is not a member of this set. We cannot write down all the integers. However, we can indicate this set by, {…,-3, -2, -1, 0, 1, 2, 3, …}, where the dots ('.') indicate the members that are not written down and *it is understood what they are.*

$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$

$\mathbb{N} = \{1, 2, 3 \ldots\}$

Notations used in writing down sets

Consider the set $\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\}$. This set can also be denoted by $\{\frac{1}{n} \mid n$ is a natural number$\}$. This is read as ' The set whose members are $\frac{1}{n}$ where n takes the values of all the natural numbers'.

$\{\frac{1}{n} : n \text{ is a positive integer}\}$

The set $\{x \mid x$ is an integer and $x>-3\}$ denotes the set whose members are x, takes all the values of integers which are greater than -3.

This set is equal to $\{-2, -1, 0, 1, 2, 3, 4, \ldots\}$.

Note:

The general situation can be described as follows: A set is determined by a defining property P of its elements, written as

$$\{x : P(x)\}$$

where $P(x)$ means that x has the property described by P. The letter x serves as a variable for objects; any other letter, or symbol except P, would have done equally well; similarly, P is a variable for properties or, as they are sometimes called, *predicates*. To avoid running into logical difficulties we shall always assume that our objects which are described by the predicate P come from a previously well defined set, say M, and sometimes we shall say so explicitly. In general then, we describe sets by

$$\{x : x \in M \text{ and } P(x)\},$$

$\{x \mid x \in M \text{ and } P(x)\}$

which also can be written as

$$\{x \in M : P(x)\}.$$

We shall use the following conventions in describing certain sets of numbers:

- $\mathbb{N}^{*} = \{0, 1, 2, 3, \ldots\}$ is the set of *natural numbers*.
- $\mathbb{N} = \{1, 2, 3, \ldots\}$ is the set of *positive natural numbers*.
- $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$ is the set of *integers*.
- $\mathbb{Q} = \{x : x = \frac{a}{b}\}$, where $a \in \mathbb{Z}, b \in Z$, and $b \neq 0$, is the set of *rational numbers*; observe that each rational number is the ratio of two integers, whence the name.
- $\mathbb{R} = \{x : x \text{ is a real number}\}$.

Exercise 1. Describe the following sets explicitly:

$$A = \{x \in \mathbb{Z} : x \lneq 5\} = \{\dots 3, 4,\}$$
$$B = \{x \in \mathbb{N} : x \text{ divides } 24\} = \{1, 2, 3, 4, 6, 8, 12, 24\} = \{3, 12, 4, 1, 24, 6, 8, 2\}$$
$$C = \{x \in \mathbb{R} : x = 2\} = \{2\}$$
$$D = \{x \in \mathbb{N}^* : x \lneq 0\} = \{\ \}$$

Exercise 2. Find a property P and a set M such that you can write the following sets in the form $\{x \in M : P(x)\}$:

$$A = \{1, 2, 4, 8, 16\} = \{x \in \mathbb{N} : x \text{ divides } 16\}$$
$$B = \{6, 4, 8, 2, 0\} =$$
$$C = \{1, 3, 5, 7, \dots\} = ?$$
$$D = \{3, 5, 11, 2, 7, 13\} = \{x \in \mathbb{N} \mid x \text{ is prime and } x \leq 13\}$$

3.1 Subset, Equality of sets, singleton set, The empty set, Universal set.

Definition: The set A is a ***subset*** of the set B if and only if every element of A is also an element of B, and denote this as $A \subseteq B$.

Note:

$\boxed{A \subseteq B}$ if and only if $\boxed{\forall x, x \in A \implies x \in B.}$ (*)

- For example,

let $A = \{1, 2, 4\}$ and $B = \{1, 2, 3, 4\}$. (Since every element of A is an element of B)

Then, A is a subset of B. i.e. $A \subseteq B$, but B is not a subset of A. i.e. $B \nsubseteq A$. (Since $\exists x_0 \in B$ such that $x_0 \notin A$)

Note:

- The relation $\subseteq$ is called the *inclusion relation*.
- $A \subseteq B$ whenever $x \in A$ implies $x \in B$.
- Observe carefully the difference between $\subseteq$ and $\in$:
- $A \nsubseteq B \iff \exists x_0, x_0 \in A \wedge x_0 \notin B$ (by *)

Lemma 1. *$\emptyset$ is a subset of every set.*

Lemma 2. *For any set A, $A \subseteq A$.*

Lemma 3. *If A is a subset of B, and B is a subset of C, then A is a subset of C.*

Exercise . Prove: If $A = B$ and $B = C$, then $A = C$.

Prove this Lemma for the general case; an example is not a proof!

Definition: A set A is a *proper subset* of a set B if $(\forall x, x \in A \Rightarrow x \in B) \wedge (\exists x_0, x_0 \in B \wedge x_0 \notin A)$.
In other words, it is not possible that A = B (whereas the symbol " ⊆ " DOES allow for that possibility).

We denote this A ⊂ B. The proper subset notation should never be used unless you are sure that equality is not possible.

Definition: Two sets A and B are ***equal*** if and only if A ⊆ B and B ⊆ A. We often use this notion to prove that two sets are equal.

Note

This can be interpreted as A = B if and only if $\forall x, x \in A \Leftrightarrow x \in B$.

I.e. $A = B$ if and only if

$[(\forall x, x \in A \Rightarrow x \in B) \wedge (\forall x, x \in B \Rightarrow x \in A)]$.

Definition: A set containing exactly one element is called a *singleton set*.

Definition: The set with no elements is called the *empty* set, denoted ∅ or { }, but not {∅}, because the last one means the set that contains the empty set.

Examples:
$\{x \lfloor x \in \mathbb{Z} \; and \; x^2 = 2\} = \emptyset$
$\{x \lfloor x \in \mathbb{R} \; and \; x^2 < 0\} = \emptyset$
Note that
$\{x \lfloor x \in \mathbb{Z} \; and \; x^2 = 2\} = \{x \lfloor x \in \mathbb{R} \; and \; x^2 < 0\}$

End of D_{13}

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Answer **all** questions in the given space.